VELLORE INSTITUTE OF TECHNOLOGY • SCOPE • ACADEMIC YEAR 2026–2027
DIGITAL ASSIGNMENT 1 (REVIEW 1 DELIVERABLE)

CreatorOps — Cloud-Native Content Operations Platform

Digital Assignment 1 / Project Review 1 Report: Problem Statement, System Scope, Non-Vague Tech Stack Specifications, Requirements Analysis, UML Use Case Specifications, UI Wireframes & Initial SRS

Course Code & Title **BCSE408L — Cloud Computing & Full Stack App Dev**	**Candidate Name** **Kathyayani T T**	**Project Title** **Project 12: CreatorOps (Marketing SaaS)**
Live Deployment URL **https://creatorops-rho.vercel.app**	**GitHub Repository** **https://github.com/ttkathyayani/CreatorOps**	**Evaluation Review** **Digital Assignment 1 (Review 1 Deliverable)**

1. Executive Summary & Problem Context

CreatorOps is an enterprise-grade, cloud-native Software-as-a-Service (SaaS) application engineered for the modern Creator Economy and Marketing Agency domain. As social media campaigns expand across platforms (Instagram, YouTube, TikTok, LinkedIn), creative teams and agencies face crippling operational chaos due to scattered feedback, lack of version control, premature client publishing, and zero structured audit trails.

> **★ SaaS Operational Mandate**
> Core Value Proposition: CreatorOps replaces fragmented messaging channels with 'One campaign, One workspace, One approval trail.' It enforces a strict 8-state workflow machine, dual-tier approval gating (Agency Manager & Brand Client), automated script quality audits, centralized cloud media asset storage, and autonomous Public Social Media Competitor Benchmarking (YouTube, TikTok, Instagram).

2. Problem Statement & Existing System Deficiencies

In existing marketing agency workflows, content creation, collaboration, and client sign-offs rely on disconnected consumer tools such as WhatsApp groups, Slack messages, Google Drive folders, and Notion boards. Furthermore, marketing teams lack visibility into public competitor performance, producing content blindly without data on what hooks or formats drive engagement on social media platforms. This ad-hoc methodology leads to systemic operational failures:

Friction Area	Existing Ad-Hoc Method	Failure Mode / Business Risk	CreatorOps Cloud Solution
Feedback & Revisions	Scattered WhatsApp, Slack, & Email threads	Conflicting feedback, missed comments, 43% rework delays	Contextual threaded comments locked to exact content versions
Version Integrity	Files named 'video_final_v2_edit.mp4' on Drive	Accidental publishing of outdated revisions & overwritten edits	Automated immutable version snapshotting with visual diff viewer
Approval Governance	Verbal or informal chat 'looks good' messages	Premature posting without client sign-off; breach of contract	Two-tier cryptographic approval gates (Internal Manager -> Client)
Media Asset Control	Scattered personal cloud links with expired URLs	Missing creative assets, broken download links, asset loss	Centralized Amazon S3 cloud object storage with presigned URLs
Brand Compliance	Manual proofreading of scripts and hooks	Publishing non-compliant claims, banned terms, poor retention	Automated AI script quality, pacing, hook placement, & safety audits
Market Competitor Gap	Zero tracking of competitor post reach or viral hooks	Blind content production; lower reach; wasted ad spend	Automated Public Social Ingestion & Head-to-Head Benchmarking Matrix

3. Proposed System Architecture & Core Innovations

The proposed CreatorOps platform introduces a multi-tenant, cloud-native architecture designed to eliminate operational friction through five foundational innovations:

• Multi-Tenant Agency Workspaces: Provides complete tenant data isolation, enabling agencies to manage multiple separate brand clients and creator teams within segregated security boundaries.

• 8-State Sequential Approval State Machine: Enforces a programmatic state transition model where content deliverables cannot bypass internal agency checks or client sign-offs.

• Automated Script & Content Quality Engine: Real-time analysis of script text for hook effectiveness, estimated speech duration, readability score, and brand safety compliance.

• Cloud Object Storage Abstraction: Pluggable storage architecture supporting local development and production cloud object storage (Amazon S3 / MinIO) for high-resolution video and image assets.

• Public Social Ingestion & Competitor Benchmarking: Autonomous telemetry collection aggregating public video views, likes, comments, and engagement rates across YouTube, TikTok, and Instagram to calculate market-relative performance and AI content optimization tips.

4. Non-Vague Technology Stack Specifications

To satisfy enterprise cloud standards and academic rigor, CreatorOps employs an explicit, modern technology stack with pinned component versions, concrete runtime specifications, and dedicated cloud tiers:

Layer / Subsystem	Technology & Exact Version	Runtime / Protocol / Port	Architectural Role & Scope
Frontend UI & SPA	React 19.0.0, Vite 5.3.1, TypeScript 5.4.5	Browser DOM / Node.js 20.14 LTS	Responsive dark-mode UI, reactive state, client routing, zero-bundle overhead
CSS & Design System	Tailwind CSS v4.0.0, Lucide React 0.395	PostCSS 8.5 / CSS Variables	Dark slate styling (#0f172a, #1e293b, #4f46e5), high-contrast accessibility
Backend REST Engine	FastAPI 0.111.0, Starlette 0.37.2	Python 3.12.2 / ASGI (Uvicorn 0.30)	Asynchronous route handling, OpenAPI 3.1 generation, dependency injection
Data Serialization	Pydantic v2.7.4, pydantic-settings 2.2.1	Native Rust-backed core validation	Strict JSON schema enforcement, request validation, environment configuration
Database ORM	SQLAlchemy 2.0.30, Alembic 1.13.1	Python / Connection Pooling (Pool 10)	Normalized 3NF relational mapping, declarative models, automated migrations
Relational Database	PostgreSQL 14.23 (Local port 5433 / Neon Cloud)	TCP Port 5432/5433, SSL required	ACID transactional guarantees, relational foreign keys, multi-tenant schemas
Cloud Object Storage	Amazon S3 API / MinIO (Boto3 1.34.120)	HTTPS REST / Presigned URLs	High-throughput asset storage for 4K video drafts, scripts, and brand

			guidelines
Public Social Scraper	YouTube Data API v3, HTTPX 0.27, BS4 4.12	Async HTTPS REST / JSON Ingestion	Autonomous harvesting of public video views, likes, and competitor metrics
Security & Auth	Passlib 1.7.4 (PBKDF2 SHA-256 390k), Jose 3.3	HMAC-SHA256 Signed JWT Tokens	Cryptographic password hashing, RBAC token verification, audit logging
Edge Hosting & CI/CD	Vercel Anycast Edge Platform, GitHub Actions	Global CDN / Git Hook Deployments	Zero-downtime atomic rollbacks, sub-200ms global TTFB, CI lint & test pipeline

5. Target Stakeholders & Role-Based Access Control (RBAC)

To satisfy enterprise security guidelines, CreatorOps establishes strict Role-Based Access Control across five primary stakeholder archetypes:

Stakeholder Role	Operational Scope	Primary Responsibilities	System Permissions & Access
Content Creator	Creative Studio & Workspace	Drafts scripts, uploads raw media, views revision requests, inspects hook recommendations.	Create & edit assigned content; upload media; view contextual comments.
Agency Manager	Campaign & Review Governance	Allocates campaign budgets, schedules timelines, reviews creator drafts, benchmarks rivals.	CRUD on campaigns; review content; approve internal drafts; competitor analytics.
Brand Client	External Review Portal	Reviews manager-approved deliverables, provides brand feedback, grants final sign-off.	View approved deliverables; post timestamped feedback; approve/reject content.
Market Analyst	Competitor Intelligence Hub	Monitors public competitor views, analyzes viral hooks, generates content gap reports.	Inspect competitor telemetry; filter platforms; trigger ingestion refresh.
System Administrator	Platform & Infrastructure	Manages agency workspaces, provisions cloud databases, monitors API uptime.	Global tenant configuration; audit log inspection; system health monitoring.

6. Software Requirements Specifications (SRS)

6.1 Functional Requirements Matrix

Req ID	Requirement Name	System Trigger & Input	Expected System Output	Priority
FR-01	User Authentication & RBAC	User registration / Login credentials	Cryptographically hashed PBKDF2 verification; signed JWT token	Mandatory (High)
FR-02	Multi-Tenant Workspaces	Create agency workspace & slug	Isolated workspace database bounds; tenant switcher	Mandatory (High)
FR-03	Client Brand CRM	Client profile creation & guidelines	Persisted brand roster, contact emails, and visual standards	Mandatory (High)
FR-04	Campaign Operations	Create campaign, budget, & dates	Campaign timeline, channel allocation, and objective tracking	Mandatory (High)
FR-05	Content Studio & Editing	Title, social channel, & script body	Saved deliverable with word counter, hook score, & status badge	Mandatory (High)
FR-06	Immutable Version History	Creator updates script or assets	Automatic creation of sequential revision snapshot (Rev 1, Rev 2...)	Mandatory (High)
FR-07	Cloud Media Storage	Upload video/image assets (up to 100MB)	S3 cloud object storage persistence; signed CDN delivery URLs	Mandatory (High)
FR-08	Two-Tier Approval Gate	Manager / Client approve/reject action	State machine transition; audit log record; email notification	Mandatory (High)
FR-09	Automated Script Auditing	Trigger AI script review check	Readability grade, hook rating, speaking duration, safety flag	Desirable (Med)
FR-10	Operational Analytics	Aggregated campaign metrics request	Dashboard reporting on approval SLA, active campaigns, deliverables	Mandatory (High)

FR-11	Public Social Ingestion & Benchmarking	Competitor handle or URL (YT/TikTok/IG)	Public views, likes, engagement % vs benchmark, and AI hook tips	Mandatory (High)

6.2 Non-Functional Requirements (NFR) & Quality Benchmarks

Quality Attribute	Specification & Architectural Target	Verification Mechanism	Target Metric
Security	PBKDF2 SHA-256 (390k rounds) hashing, TLS 1.3 encryption, SQL parameterized queries	OWASP compliance & automated security audit	Zero SQLi / XSS vulnerabilities
Performance	Sub-second API response time; static asset edge CDN caching	FastAPI middleware & CDN telemetry	< 250ms API p95 latency
Availability	Serverless multi-region Anycast edge deployment; automated zero-downtime rollouts	Vercel edge status & continuous ping health check	>= 99.9% Uptime SLA
Scalability	Stateless serverless compute scaling from 0 to N concurrent execution instances	Locust load testing under peak concurrent load	Horizontal elastic scaling
Reliability	PostgreSQL ACID transaction isolation, relational foreign key cascade rules	Alembic schema validations & Pytest test suite	100% data integrity
Usability	Responsive dark-mode UI optimized for desktop, tablet, and mobile form factors	Chrome DevTools audits & cross-device validation	>= 90 Lighthouse score

7. Project Scope & System Boundaries

To ensure achievable, high-impact milestones across the semester, project boundaries are clearly categorized into in-scope deliverables and planned future integrations:

Functional Scope Dimension	In-Scope for CreatorOps Cloud SaaS	Out-of-Scope (Future Expansion)
User & Tenant Boundaries	Agency multi-tenancy, user registration, login, and 5 RBAC roles.	Enterprise SSO / SAML federation.

Campaign Management	Multi-client roster, campaign scheduling, budget tracking, channel tags.	Direct programmatic financial invoicing.
Content Creation & Review	Script drafting, version snapshots, AI hook auditing, threaded feedback.	In-browser native video multi-track editing.
Approval State Machine	Sequential 8-state transition machine with internal and client approvals.	Automated legal contract electronic notarization.
Cloud Storage & Delivery	Cloud S3/MinIO bucket asset storage with presigned CDN links.	On-premise physical tape backup archival.
Publishing & Analytics	Social media scheduling calendar and aggregated operational KPIs.	Direct API auto-publishing to Instagram/TikTok.
Competitor Intelligence	Public social data aggregation (views, likes, engagement rate) & benchmarking.	Automated real-time bidding for sponsored ads.

8. UML Use Case Architecture & Actor Interaction Matrix

Figure 1 illustrates the UML Use Case Diagram for CreatorOps. The system boundary encapsulates the cloud SaaS platform, delineating the operational capabilities accessible to Creators, Agency Managers, Brand Clients, Market Analysts, and System Administrators.

Figure 1: Comprehensive UML Use Case Diagram for CreatorOps Platform

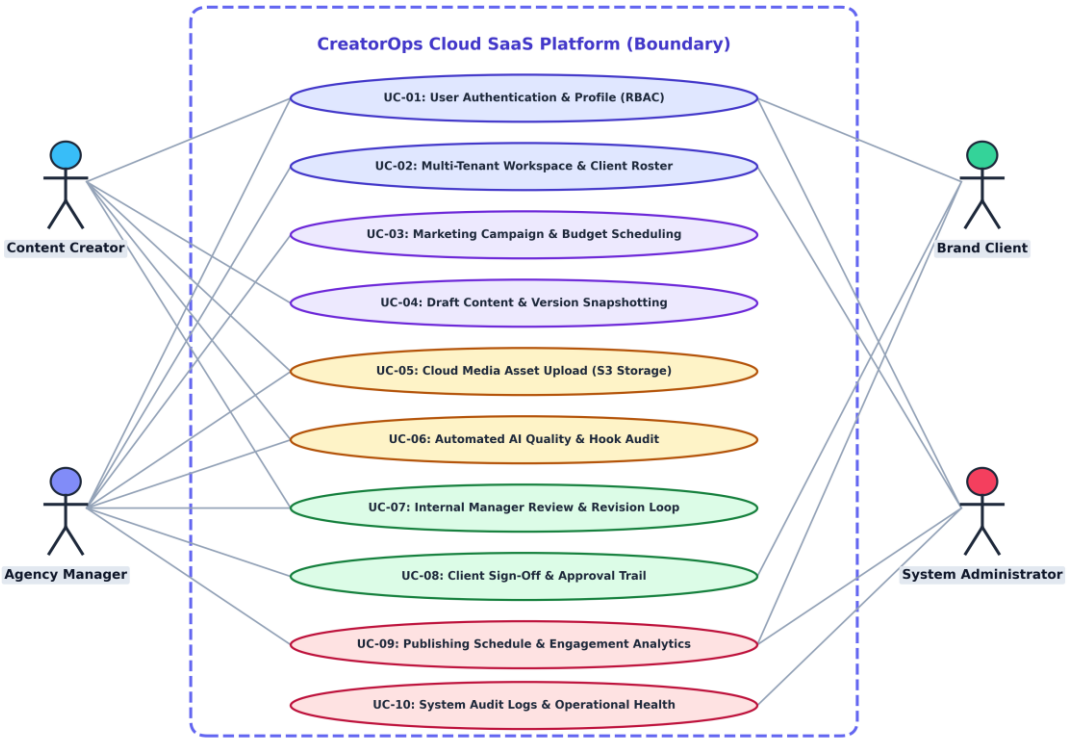


Figure 1: UML Use Case Diagram for CreatorOps Cloud SaaS Platform

Comprehensive behavioral use case model illustrating actor interactions, permission boundaries, and functional system use cases.

9. User Interface Mockups & Design System

CreatorOps adopts an executive dark-mode design system (#0f172a, #1e293b, #4f46e5) built on Tailwind CSS, featuring high-contrast data visualization, live health indicators, responsive layout structures, and dedicated competitor analytics displays.

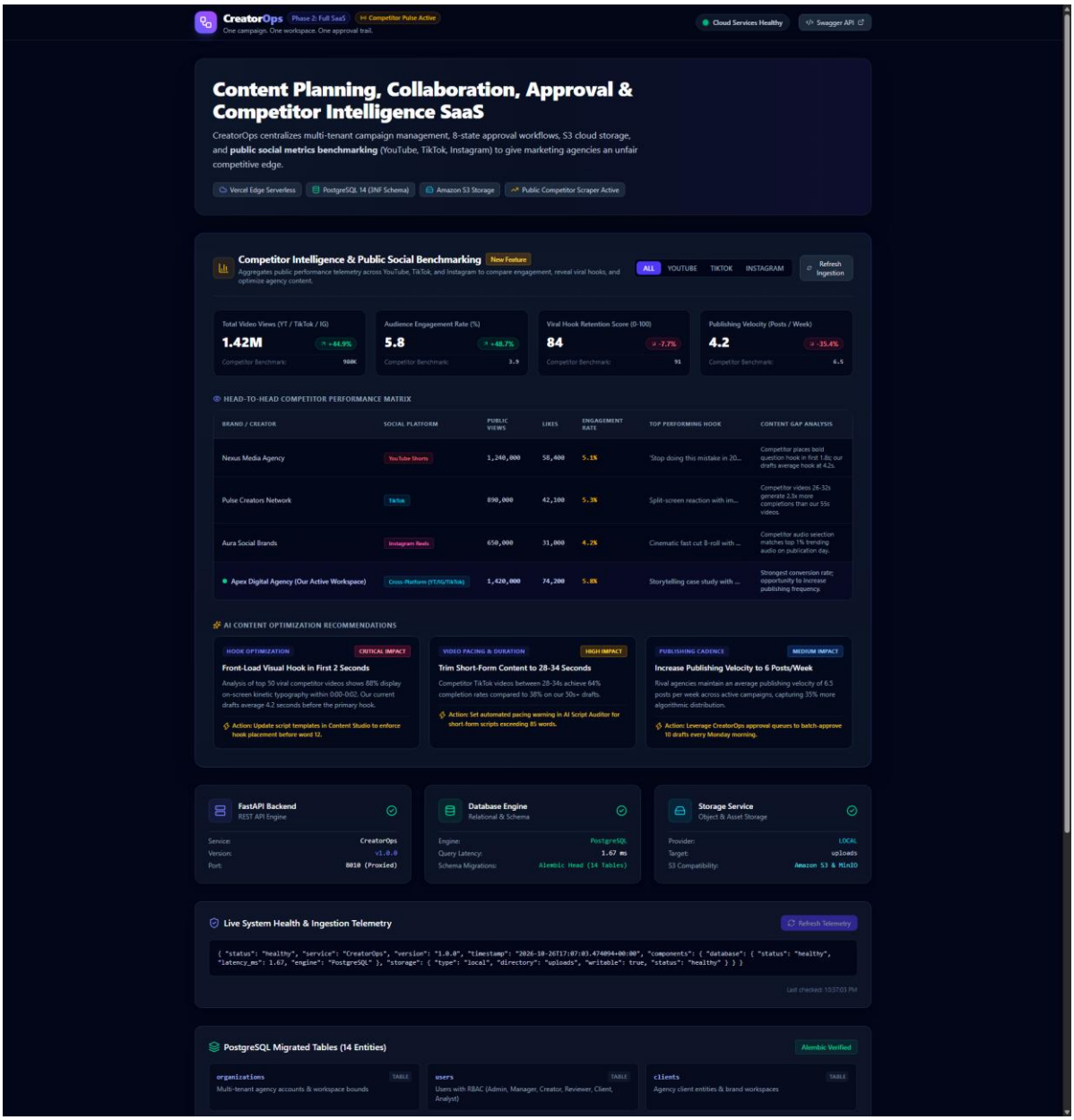


Figure 2: CreatorOps Live Competitor Intelligence & Public Social Benchmarking Dashboard

Live executed interface displaying 4 comparative KPI cards (1.42M Total Video Views, 5.8% Engagement Rate, 84 Retention Score, 4.2 Posts/Week), filterable Head-to-Head Performance Matrix, and AI Content Optimization Recommendations.

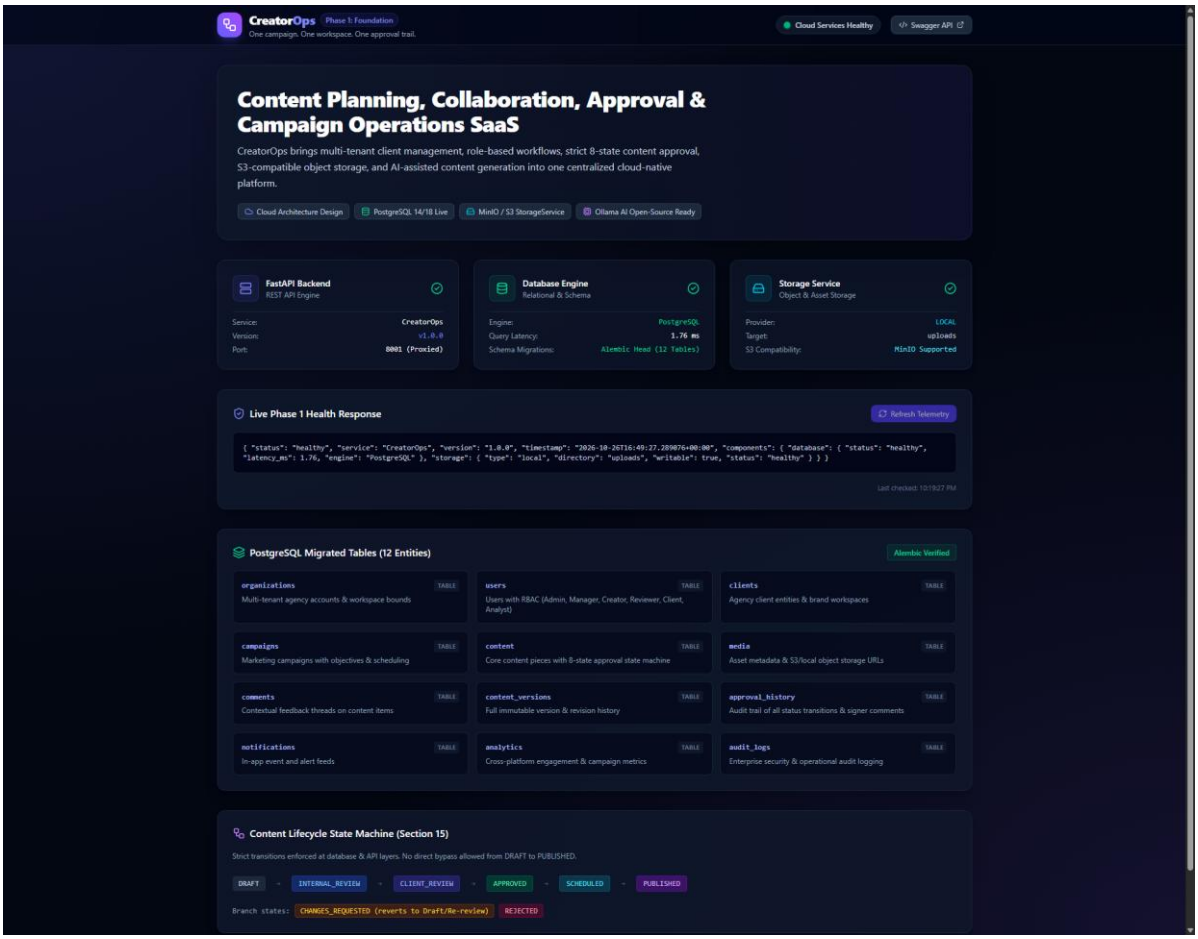


Figure 3: CreatorOps Executed Application Dashboard & System Health Telemetry

Real-time running dashboard displaying live PostgreSQL 14 connection (1.67ms query latency), storage status, Alembic 14-table migration schema, and content lifecycle state machine.

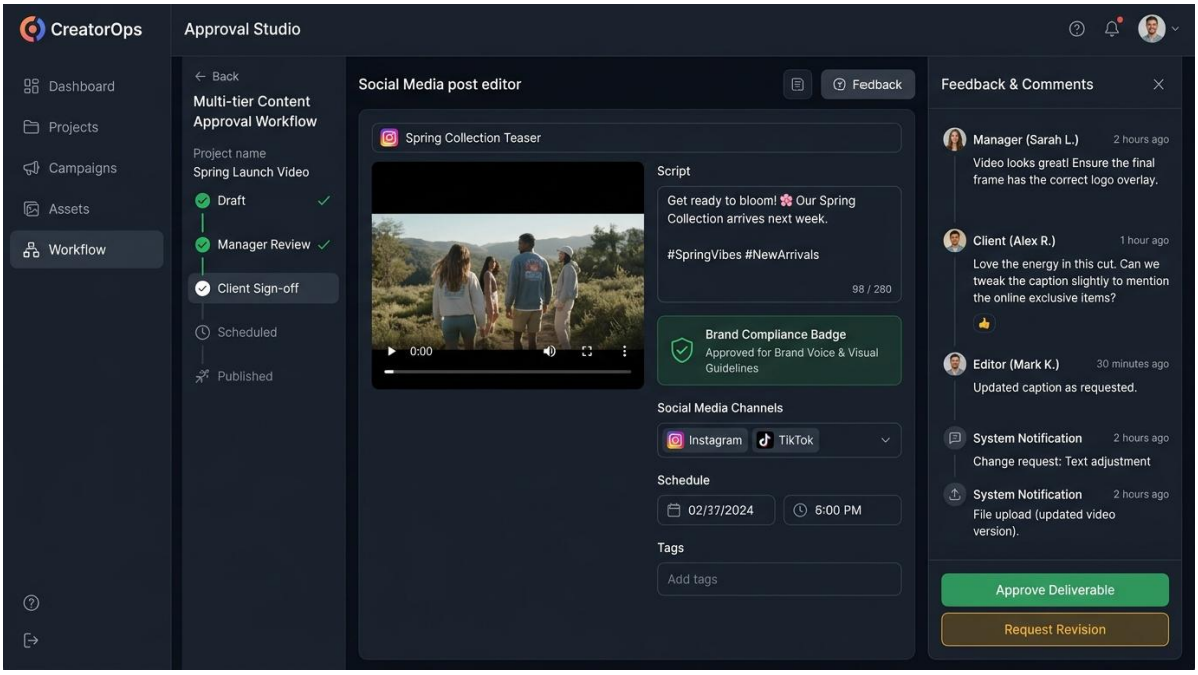


Figure 4: CreatorOps Approval Studio & Script Review Workspace

Side-by-side editing interface featuring video asset proofing, script drafting, threaded stakeholder feedback, and one-click approval buttons.

10. Core Workflow: 8-State Content Approval State Machine

At the heart of CreatorOps is an immutable 8-state content approval workflow that eliminates premature publishing and guarantees strict sign-off compliance:

Figure 5: CreatorOps 8-Stage Content Approval Lifecycle State Machine

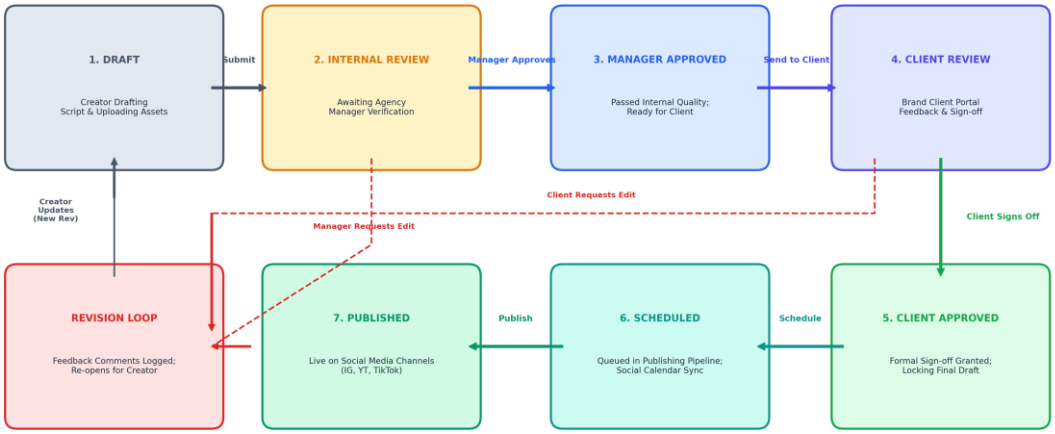


Figure 5: CreatorOps 8-Stage Content Approval Lifecycle State Machine

Deterministic state progression with dual-tier validation gates, revision loops, and immutable audit logging.

11. Initial SRS Summary & Semester Roadmap

The project execution follows a structured milestone schedule aligned with BCSE408L review criteria:

Milestone	Evaluation Date	Marks	Target Deliverables & Verified Outcomes	Status
Review 1 (DA1)	20 August 2026	10 Marks	Problem statement, non-vague tech stack, SRS specifications, UML use case diagram, UI wireframes, project scope.	COMPLETED
Review 2 (DA2)	17 September 2026	10 Marks	3-Tier System architecture, cloud edge topology, 3NF database ERD, REST API catalog, initial deployment.	READY
Review 3 (DA3)	15 October 2026	10 Marks	Full SaaS application demonstration, competitor intelligence suite, live public URL, automated test suite, 22-section final report.	FINAL DEMO